PHYS 253: Thermal Physics

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Abstract

Lecture notes for PHYS 253 (Thermal Physics) at McGill University, Fall 2026. The notes follow the lectures and are cross-referenced with *An Introduction to Thermal Physics* by Daniel V. Schroeder. Dates in the margin mark the lecture each block of material came from. These notes are a work in progress.

Contents

1 Introduction

1.1 Course Information

The textbook is *An Introduction to Thermal Physics* by Daniel V. Schroeder. The course is taught by Bradley Siwick, Mondays, Wednesdays and Fridays from 8:35 to 9:25 in Rutherford 118. Grading is divided as follows:

- 10% problem sets ($\sim$ 6, submitted as handwritten PDF scans)[1]
- 50% two midterms
- 40% final exam

August 31, 2026

1.2 What Thermodynamics Is

Definition 1 (Thermodynamics). *Thermodynamics, from the ancient Greek therme (heat) and dynamis (power), is the branch of physics that deals with the relationships between heat, work, temperature and energy. It studies the equilibrium properties of large-scale systems in which temperature is an important variable.*

Remark 1. *Thermodynamics does not describe the microscopic details of a system[2]. It describes the macroscopic properties and behaviour of a system using a small number of state variables.*

2 Equilibrium and State Variables

September 2, 2026

2.1 Equilibrium States

Definition 2 (Equilibrium state). *A system is in an equilibrium state if all the measurable bulk physical properties of the system are uniform throughout the system and constant in time.*

Remark 2. *For a fixed volume of gas in an equilibrium state, the pressure is the same everywhere in the gas and the volume does not change in time.*

Remark 3. *The pressure, volume and mass of a gas fully describe its equilibrium state. Once those are fixed, every other measurable bulk property is fixed as well.*

2.2 Equations of State

Definition 3 (Equation of state). *An equation of state is a functional relationship between the easily measured state variables[3] of a system in an equilibrium state.*

[1] Only one or two problems on each assignment are marked for submission and grading.
[2] Statistical mechanics and more recent extensions such as stochastic thermodynamics, do.
[3] Such as pressure, volume and temperature.

Example. *The ideal gas law is an equation of state relating* (P, V, T)*. It applies to equilibrium states of dilute gases at high temperature:*

$$PV = nRT,$$

where R *is the ideal gas constant.*

2.3 State Functions

Definition 4 (State function)**.** *A state function is a property of a system that depends only on the current equilibrium state, and not on the path taken to reach that state. Internal energy, enthalpy, entropy and Gibbs free energy are all state functions.*

Remark 4. *State functions matter because their values can be compared between two equilibrium states without knowing anything about the process that connected them.*

Example. *Temperature is a state function. It is the property that determines whether two systems are in thermal equilibrium*[4] *with each other.*

3 Laws of Thermodynamics

3.1 Zeroth Law

Law 1 (Zeroth Law of Thermodynamics)**.** *If two systems A and B are each separately in thermal equilibrium with a third system C, meaning no heat flows between A and C and no heat flows between B and C, then A and B are in thermal equilibrium with each other.*

Remark 5. *The zeroth law is what makes a thermometer work. System C is the thermometer, and the law guarantees that two systems reading the same value on it are in thermal equilibrium with each other.*

Law 2 (First Law of Thermodynamics)**.** *The relationship between Heat (Q), Work (W) and the Internal Energy (U) of a system is dictated by the conservation of energy.*

4 Energy, Work and Heat

September 4, 2026

4.1 Energy

Definition 5 (Energy)**.** *Energy, from the Greek energeia meaning "in work," is the capacity to do work. It is a property of a thermodynamic system, and it is stored in different forms depending on the phase and composition of the matter involved.*

[4] Thermal equilibrium only, not full thermodynamic equilibrium.

4.1.1 Forms of Energy

Definition 6 (Kinetic energy). *From the Greek kinesis (motion), the energy associated with the motion of a body or of the particles inside it.*

Definition 7 (Potential energy). *Energy due to condition, position or composition. It is associated with interactions, meaning forces of attraction or repulsion between objects, or chemical bonding.*

4.2 Work

Definition 8 (Work). *Work is done when a force acts through a distance, and it is associated with a change in the energy of a system:*

$$W = \int \vec{F} \cdot d\vec{r}.$$

In thermodynamics, work is any transfer of energy into or out of a system that is not heat.

Compression of a Gas by a Piston

Take a gas in a cylinder closed by a piston of area A. Pushing the piston in by a small distance δx does work

$$W = F\,\delta x.$$

The pressure of the gas is $P = F/A$, so $F = PA$ and

$$W = PA\,\delta x.$$

For a quasistatic compression[5], the change in volume is $\delta V = -A\,\delta x$, so the work done *on* the gas is

$$W = -P\,\delta V \qquad \text{(quasistatic)}.$$

Compressing the gas means $\delta V < 0$, so $W > 0$ and the energy of the gas goes up, as expected.

Remark 6. *Energy and work are measured in the same units, joules, where* $1\,\mathrm{J} = 1\,\mathrm{N\,m} = 1\,\mathrm{kg\,m^2\,s^{-2}}$.

4.3 Heat

Definition 9 (Heat). *Heat is energy that flows between systems because of a temperature difference, and it drives them toward thermal equilibrium. It is the quantity of energy q transferred between a system and its surroundings as a result of that temperature difference.*

[5] Slow enough that the gas stays in internal equilibrium at every instant, so the pressure is uniform throughout the volume and the ideal gas law still applies.

Remark 7. *Heat and work are not properties of a thermodynamic system. They are the two ways energy can be exchanged between a system and its surroundings. A system does not "contain" heat or work, it contains energy.*

5 Misc

September 9, 2026

Application of the first law of thermodynamics

$$\Delta U = Q + W$$

Compression work $W = -P\Delta V$ is a deceptively simple expression. For an ideal gas $PV = nRT$, so P and V and not independent state-variables. There are 2 limiting cases for the compression/expansion of a gas:

- Isothermal: T is constant during the process.
- Adiabatic: No exchange of heat during the process; no heat flow through boundary.

Quasistatic Isothermal compression/expansion I: Work $PV = nRT = Nk_BT$ equation of state always satisfied during process (Note $N = nN_A; R/N_A = k_B = 1.38 \times 10^{-23} J/K$). $W = -NkT \int_{V_i}^{V_f} \frac{1}{V} dV = NkT \ln(\frac{V_i}{V_f})$. Quasisttic Isothermal Compression/Expansion II: Heat. $\Delta U = Q + W$. Remember, for an ideal gas $U(T) = N\bar{e_k} = \frac{3Nk_BT}{2}$(For a monatomic gas) $= \frac{fNk_BT}{2}$(For a more complex gases). For isothermal compression/exapansion $\Delta U = 0$, so $Q = -W = NkT \ln(\frac{V_f}{V_i})$. Quasistatic Adiabatic Compression/Expansion: No Heat. $\Delta U = Q + W = 0 + W = W$ for an adiabatic process. Gas temperature must change $U(T)$. Equation for the Adiabat $U = \frac{f}{2}NkT$. $dU = f/2NkdT = -PdV = dW$, plugging in $f/2dT/T = -dV/V$, integrate $f/2ln(T_f/T_i) = -ln(V_f/V_i)$. Exp: $V_fT_f^{f/2} = V_iT_i^{f/2}$, where $VT^{f/2}$ is a constant. Elim. T using IGL: $V^\gamma P$ = constant, where γ is called the adiabatic exponent, is an abbrevitation for $(f+2)/f$.

A Appendix

B Useful Links